



Final Project Data Analytics

Arie Trisnasaputra - Batch 31





INTRODUCTION



Arie Trisnasaputra



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Education:

- **Bina Nusantara University (2019-2024)**
Business Creation Major
Bachelor Economics
- **Dibimbing.id (2025)**
Data Science / Analytics Bootcamp

Experience:

- **IT Project Manager and Finance Intern (2024)**
PT Inacash Lentera Teknologi
- **Teaching Assistant (2020-2021)**
Management Laboratory BINUS



Overview Project

Project Overview

This project analyzes a beverage company's sales and focus on Product Performance and Discount Analysis.

Business Objectives:

1. Find out the best selling beverage category in each region
2. Which product generates most revenue
3. Does discount affect to the quantity sold
4. Which discount rate are the most effective





Main Project

Dataset

Order_ID	Customer_ID	Customer_Type	Product	Category	Unit_Price	Quantity	Discount	Total_Price	Region	Order_Date
ORD956181	CUS116	B2C	Apollinaris	Water	99	6	0,00	594	Brandenburg	01 May 2023
ORD226366	CUS4840	B2C	Mountain Dew	Soft Drinks	99	2	0,00	198	Sachsen-Anhalt	14 December 2023
ORD2618213	CUS4912	B2C	Kölsch	Alcoholic Beverages	99	6	0,00	594	Schleswig-Holstein	12 July 2023
ORD1184604	CUS4410	B2C	Kölsch	Alcoholic Beverages	99	5	0,00	495	Mecklenburg-Vorpommern	14 September 2023
ORD2146748	CUS3705	B2C	Apollinaris	Water	99	5	0,00	495	Berlin	11 July 2023
ORD561480	CUS6591	B2C	Mountain Dew	Soft Drinks	99	7	0,00	693	Brandenburg	24 May 2023
ORD749593	CUS3698	B2C	Apollinaris	Water	99	4	0,00	396	Baden-Württemberg	25 March 2023
ORD1315992	CUS8702	B2C	Mountain Dew	Soft Drinks	99	14	0,00	1386	Saarland	21 February 2023
ORD1508532	CUS7644	B2C	Kölsch	Alcoholic Beverages	99	3	0,00	297	Baden-Württemberg	24 May 2023
ORD14667	CUS37	B2C	Kölsch	Alcoholic Beverages	99	5	0,00	495	Rheinland-Pfalz	05 April 2023
ORD1259663	CUS57	B2C	Apollinaris	Water	99	3	0,00	297	Mecklenburg-Vorpommern	09 May 2023
ORD2293887	CUS4915	B2C	Kölsch	Alcoholic Beverages	99	15	0,00	1485	Brandenburg	25 February 2023
ORD585481	CUS1662	B2C	Mountain Dew	Soft Drinks	99	9	0,00	891	Mecklenburg-Vorpommern	24 September 2023
ORD2379810	CUS9067	B2C	Mountain Dew	Soft Drinks	99	6	0,00	594	Bremen	17 March 2023
ORD2354418	CUS3557	B2C	Mountain Dew	Soft Drinks	99	13	0,00	1287	Thüringen	01 December 2023
ORD1130836	CUS5315	B2C	Mountain Dew	Soft Drinks	99	14	0,00	1386	Hessen	10 September 2023

Dataset contains 8999910 rows and 11 columns

Link to dataset : <https://www.kaggle.com/datasets/sebastianwillmann/beverage-sales>

Data Preprocessing

Random Sampling

Because the data contains large amount of rows and it is not resource efficient to process, we took random sampling using slovin method to calculate the amount of sample that can represent the original data.

Sample size : 9989

Data Cleaning

Findings from data cleaning:

1. No missing / incorrect values
2. No duplicates
3. No outlier values detected because the values on quantity and total_price still makes sense

Data Manipulation

1. Because the data type of order_date is object, we transform the data type into datetime
2. Split the datetime into year, month, and day of week.

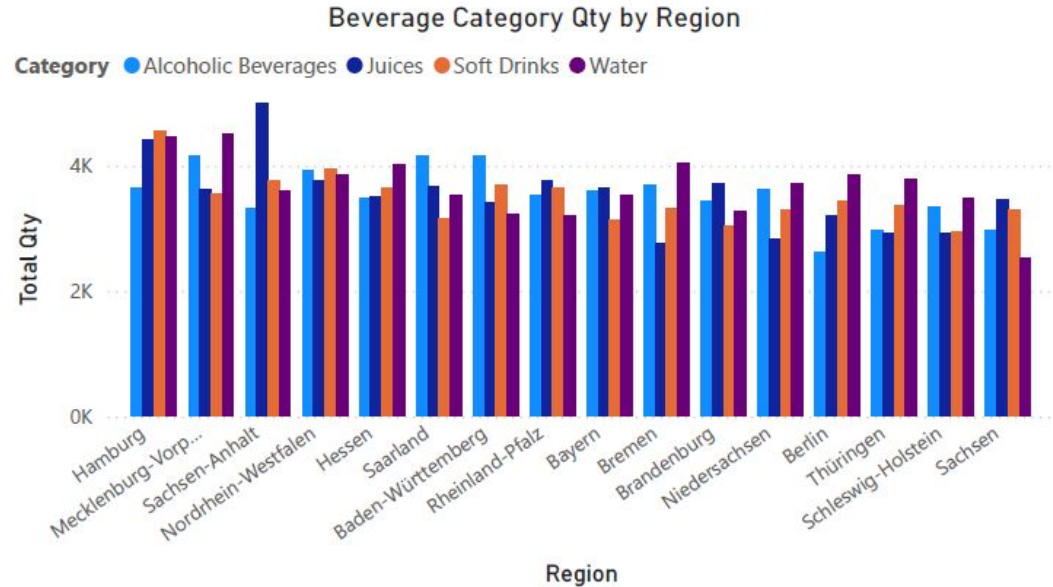


Findings and Results

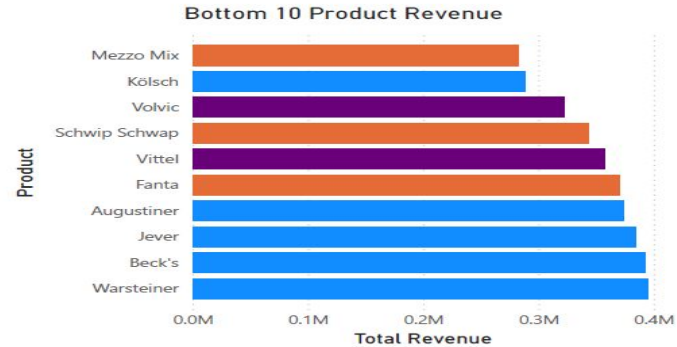
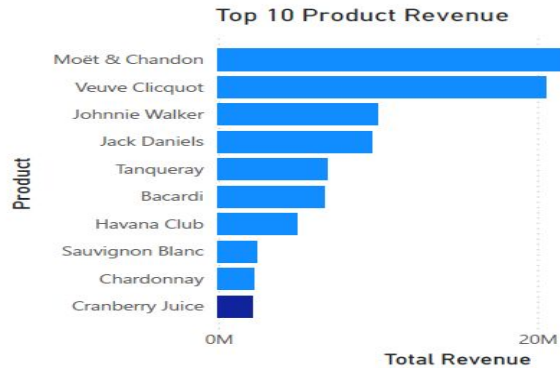
1. Best Selling Beverage Category in Each Region

Insights:

1. Alcoholic Beverages are strong in Baden-Wurttemberg, Saarland, and Mecklenburg-Vorpommern and weak in Berlin, Sachsen, and Thuringen.
2. Soft drinks have a pretty balanced performance but exceptionally in Hamburg
3. Juices in Sachsen-Anhalt is the top quantity sold above any other region and any other category



2. Which Product Generates Most Revenue



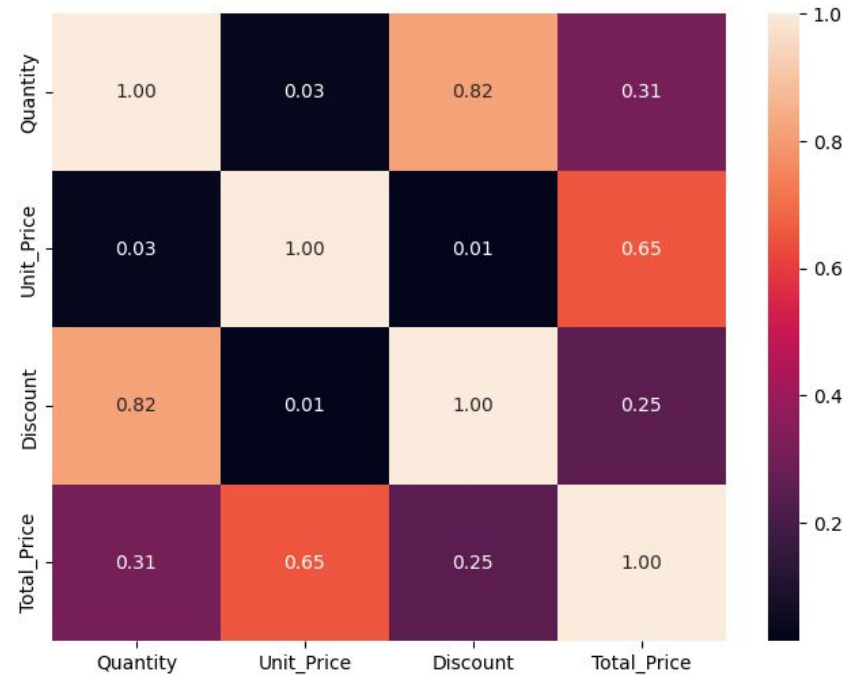
Insights:

1. Top 2 product total revenue is close to each other and dominates
2. Bottom 10 product can be used as a bundle for promotions

3. Does the discount affect the quantity sold

Insights:

Based on the correlation heatmap on the side, Discount and Quantity have strong correlation with 0.82 score. Which means discount positively impact on the quantity of product sold.



3. Does the discount affect the quantity sold cont.

To analyze further we use ANOVA analysis to find are there any significant differences in discounts and normal.

Based on the results on the side H0 rejected and that means there are significant differences in discounts and normal. This analysis is to prove that discount does have impact on quantity sold.

```
from scipy.stats import f_oneway

# ANOVA
f_stat, p_value = f_oneway(discount0, discount1, discount2, discount3)
print(f"F-Statistic: {f_stat}, P-Value: {p_value}")

# Keputusan
alpha = 0.05
if p_value <= alpha:
    print("Reject H0, there are significant differences")
else:
    print("H0 Accepted, there are no significant differences")

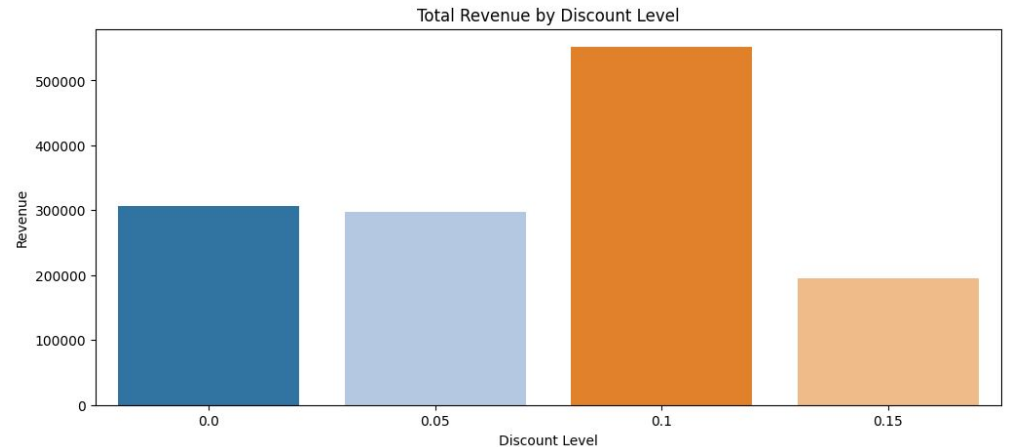
✓ 0.0s

F-Statistic: 39.93600010076588, P-Value: 6.677340816943314e-20
Reject H0, there are significant differences
```

4. Which Discount are The Most Effective

Insights:

1. Based on the graph alone, 10 % discount have the most total revenue above any other discount and normal price.
2. 15% percent discount on the other hand, have a negative impact on the revenue where the total revenue is more less than normal price.



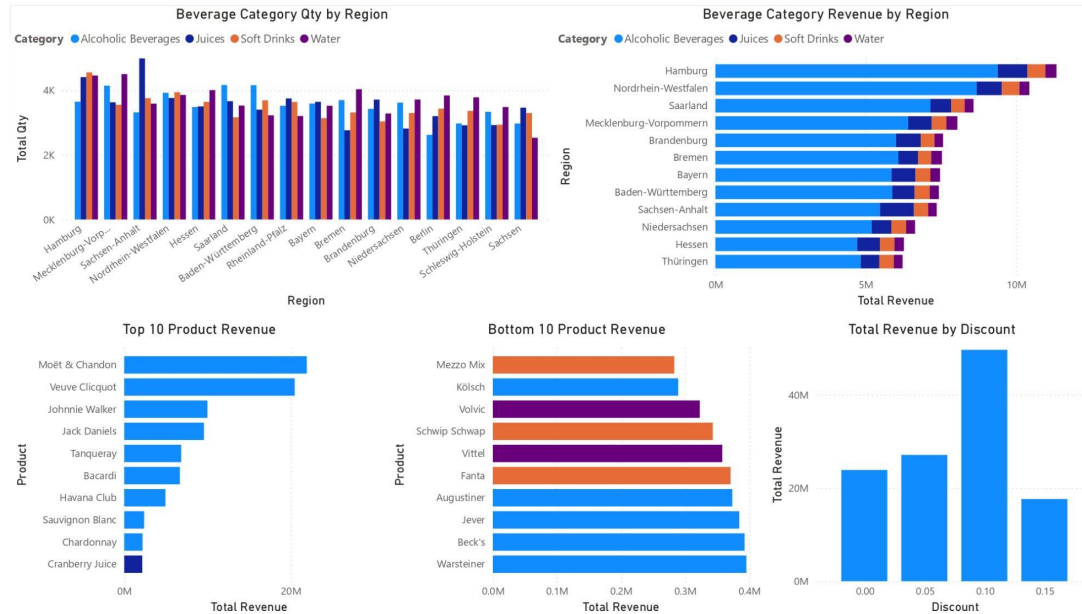
Summary

1. Alcohol Beverages dominate the quantity sold especially in the Baden-Wurttemberg, Saarland, and Mecklenburg-Vorpommern. Soft drinks and juices are relatively consistent in sales
2. Moët & Chandon and Veuve Clicquot dominate product total revenue and almost all top revenue are from alcoholic beverages
3. Discount does affect on the quantity sold strongly supported by ANOVA F stats $39 > \alpha 0.05$ that shows there are significant differences variance on discount and normal price
4. 10 % discount does impact heavily on revenue, while on the other hand 15% discount gave negative impact on revenue than normal price

Recommendations

1. Keep utilizing the 10% discount because it is proven that 10% have most impact on revenue, and reconsider about 15%+ discount because it shows of drop on revenue
2. Focus future promotions on region product pairs because each region have their own preferences on beverage category
3. Premium alcohol brand (Moët & Chandon and Veuve Clicquot) are key revenue drivers should be more prioritized on the future campaign
4. For bottom revenue product can be used as a bundle for promotion

Dashboard



Thank You



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